

1lbm unsalted peanuts

$\frac{3}{4}$ tsp NaCl

1 tsp salt = 6g

$$6 \cdot \frac{3}{4} = 4.5 \text{ g NaCl in final mixture}$$

$$75 \text{ mg Na/oz} - 5 \text{ mg Na/oz} = 70 \text{ mg Na/oz}$$

$$\frac{70 \text{ mg Na}}{\text{oz}} \cdot \frac{1 \text{ oz}}{1 \text{ lbm}} \cdot \frac{1 \text{ g}}{1000 \text{ mg}} = 1.12 \text{ g Na/lbm}$$

$$\begin{array}{l} \text{Na: } 22.99 \text{ g/mol} \\ \text{Cl: } 35.45 \text{ g/mol} \end{array} \quad \frac{22.99}{22.99 + 35.45} = 0.393 \cdot 100 = 39.3\%$$

75 mg of Na in 1 oz of peanuts + salt

$$1.12 = 0.393 \cdot \text{NaCl}$$

$$\text{NaCl} = 2.85 \text{ g}$$

$$4.5 \text{ g} - 2.85 \text{ g} = 1.65$$

$$1.65 \text{ g} \cdot \frac{1 \text{ tsp}}{6 \text{ g}} = \boxed{0.275}$$

1. a.) 0.275 teaspoons of salt



$$150 - 5 = 145 \text{ mg Na} / \text{oz}$$

$$\frac{145 \text{ mg Na}}{\text{oz}} \cdot \frac{16 \text{ oz}}{1 \text{ lbm}} \cdot \frac{1 \text{ g}}{1000 \text{ mg}} = 2.32 \text{ g Na} / \text{lbm}$$

$$2.32 = 0.393 \cdot \text{NaCl}$$

$$\text{NaCl} = 2.32 / 0.393 = 5.90 \text{ g NaCl}$$

$$4.5 \text{ g} - 5.9 \text{ g} = -1.4 \text{ g}$$

$$1.4 \text{ g} \cdot \frac{1 \text{ tsp}}{6 \text{ g}} = 0.23 \text{ tsp}$$

1.b.) you must extract 0.23 teaspoons of salt from the salted peanuts somehow

1.c.) you cannot make Professor Wedge wood's peanut butter recipe with salted peanuts and normal means.

Pr. 1: 30/30

100g Cu $\downarrow$ Cl_2

$$\text{CuCl}_2 + \text{CuCl} = 200\text{g}$$

$$\text{Cu: } 63.546 \text{ g/mol}$$

$$\text{Cl: } 35.453 \text{ g/mol}$$

$$\text{CuCl}_2 = 63.546 + 2 \cdot 35.453 = 134.452 \text{ g/mol}$$

$$\text{CuCl} = 63.546 + 35.453 = 98.999 \text{ g/mol}$$

~~100g~~

$$\text{CuCl}_2 \text{ Cu frac.} = \frac{63.546}{134.452} = 0.473$$

$$\text{CuCl Cu frac} = \frac{63.546}{98.999} = 0.642$$

$$0.473x + 0.642y = 100$$

$$x + y = 200$$

$$y = 200 - x$$

$$100 = 0.473x + 0.642(200 - x) \rightarrow -0.169x = -128.4, x = 168.05$$

$$y = 200 - 168.05 = 31.95$$

$$168.05 / 31.95 = 5.26$$

2.) 5.26 g of CuCl_2 per 1g of CuCl

$$C_p(T) = A + BT + CT^2 + DT^3$$

$$A = 28.11 \quad B = -3.86 \text{ E-6} \quad C = 1.746 \text{ E-5} \quad D = -1.065 \text{ E-8}$$

~~$$C_p(300) = 28.114 \text{ J/mol}\cdot\text{K}$$~~
~~$$C_p(700) = 28.120 \text{ J/mol}\cdot\text{K}$$~~

$$\frac{dH}{dT} = C_p \quad \int dH = \int C_p dt$$

$$\Delta H = \int C_p dt$$

$$\Delta H = \int (28.11 + -3.86 \text{ E-6} T + 1.746 \text{ E-5} T^2 - 1.065 \text{ E-8} T^3) dt$$

$$= 28.11T - \frac{3.86 \cdot 10^{-6} T^2}{2} + \frac{1.746 \cdot 10^{-5} T^3}{3} - \frac{1.065 \cdot 10^{-8} T^4}{4} \quad \left| \begin{array}{l} 700 = T_2 \\ 300 = T_1 \end{array} \right.$$

$$\boxed{3a.) \Delta H = 12464 \text{ J/mol}}$$

Pr. 3:
15/20

~~$$2.0 \quad \frac{700}{1000} = 0.7 \text{ kK} \quad \frac{300}{1000} = 0.3 \text{ kK}$$~~

$$28.11T - \frac{3.86 \cdot 10^{-6} T^2}{2} + \frac{1.746 \cdot 10^{-5} T^3}{3} \quad \left| \begin{array}{l} 0.7 = T_2 \\ 0.3 = T_1 \end{array} \right. = 11.244$$

$$\boxed{b.) \Delta H = 11244 \text{ J/mol}}$$

$$11.244 \cdot 1000 = 11244$$

Both should be same, -5 pts

$$\frac{1}{40-9\sqrt{x-1}} \rightarrow \frac{1}{4} \cdot \frac{1}{10-\sqrt{x-1}} \quad \begin{aligned} &u = \sqrt{x-1} \\ &u^2 = x-1 \\ &u^2 + 1 = x \\ &2u du = dx \end{aligned}$$

$$\frac{1}{4} \cdot \frac{1}{10-u} \cdot 2u du = \frac{1}{2} \cdot \frac{u}{10-u} du$$

$$\frac{1}{2} \cdot \left(-\frac{10-u}{10-u} + \frac{10}{10-u} \right) du \quad \frac{1}{2} \cdot \frac{u+10-10}{10-u}$$

$$\int \frac{1}{2} \cdot \left(\frac{10}{10-u} - 1 \right) du = \frac{1}{2} \cdot \left(-10 \ln|10-u| - u \right) \Big|_1^h$$

$$\frac{1}{2} \left(-\sqrt{x-1} - 10 \ln|10-\sqrt{x-1}| \right) \Big|_1^h$$

$$\frac{1}{2} \left(-\sqrt{h-1} - 10 \ln|10-\sqrt{h-1}| + \sqrt{1-1} + 10 \ln|10-\sqrt{1-1}| \right)$$

$$4.) = \left[-\frac{\sqrt{h-1}}{2} - 5 \ln|10-\sqrt{h-1}| + 5 \ln|10| \right]$$

ethanol 7 kcal/g $\Delta H^\circ_{\text{gas}} = -1235 \text{ kJ/mol}$

STP

$$\Delta H^\circ_{\text{liq}} = -1367 \text{ kJ/mol}$$

~~7 kcal/g~~. ~~ethanol~~ ethanol: $\text{CH}_3\text{CH}_2\text{OH}$

molar mass =

$$2 \cdot 12.011 + 6 \cdot 1.0078 + 15.999 = 46.07 \text{ g/mol}$$

$$7 \frac{\text{kcal}}{\text{g}} \cdot \frac{46.07 \text{ g}}{1 \text{ mol EtOH}} \cdot \frac{4184 \text{ J}}{\text{kcal}} \cdot \frac{1 \text{ kJ}}{1000 \text{ J}} = 1552.47 \text{ kJ/mol}$$

(5.1)

$$1552.47 > 1235$$

$$1552.47 > 1367$$

Pr. 5:

10/10

$\therefore$ Google gives an inaccurate/approximate value for ethanol in kcal/g. Additionally, it most likely gives a value for ethanol in the liquid state.

There is a minus sign for the standard enthalpies because combustion of ethanol exerts energy onto the surroundings, rather than absorbing it.